Complete finite Mumford census, F3 and F5

All four frozen cells complete. Rust and separately implemented C++ agree on every valid pair, actual Cantor addition, group table and cyclic subgroup. Each field's two fresh-process repeats are byte-identical in six deterministic result files. This checks determinism on one host; scientific PCC standing remains unsigned/unverified. [pcc:unverified]

Exact value, per repeat	F3	F5
Monic quintics attempted	243	3125
Squarefree / singular	162 / 81	2500 / 625
Actual Cantor additions checked	31,560	2,707,800
Associativity triples checked	595,488	111,960,960
Distinct cyclic subgroups across models	708	16,624
Proper nontrivial unequal fractions	321	11,005
Odd-prime, noncharacteristic unequal stratum	42	1510

Every monic degree-five polynomial is included, with singular rejections retained. Models are weighted by their coefficient code, not by isomorphism class. Distinct subgroups are deduplicated by membership within each model. Smoothness means split u; identity is included. Exact cross-products compare each subgroup fraction with its ambient fraction. No model or subgroup was selected after viewing its result. [pcc:unverified]

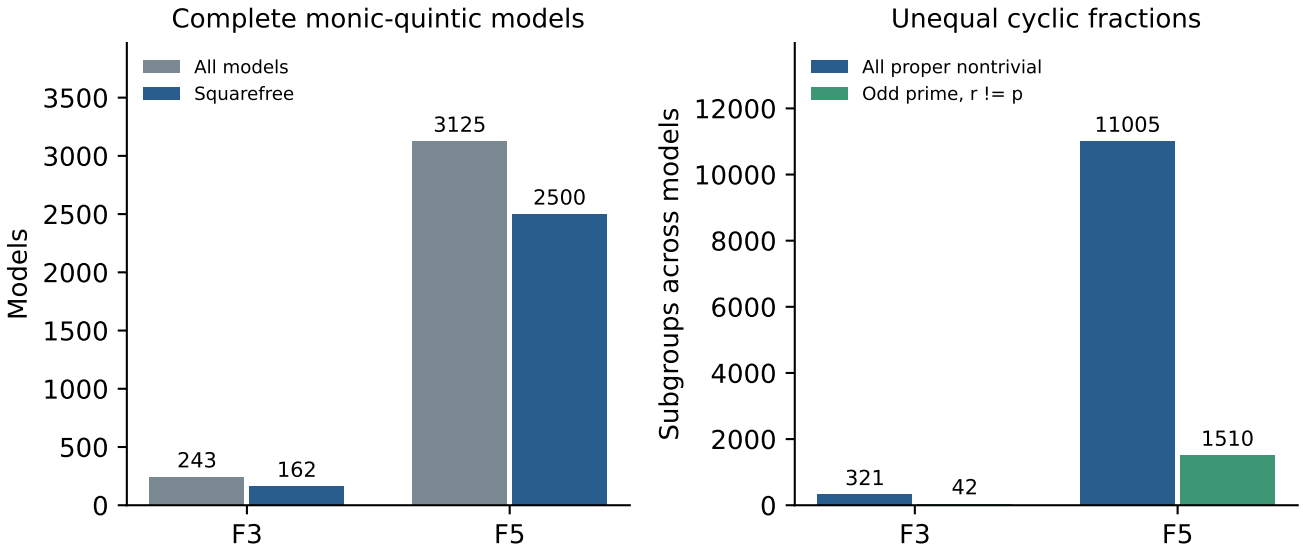


Figure 1: Recorded r0 counters. The r1 results match byte-for-byte. Complete exact populations have no sampling interval; no interfield unusualness inference is made.

Pre-data controls pass: $x^5 - x$ has 8/7 pairs/smooth over F3 and 16/16 over F5. The source-known Stoll $x^5 + 1$/F3 control has 10/8 with the proper order-five cycle generated by $(X, 1)$ all smooth. The latter's five-cycle arithmetic is separately formalized in Lean; full ambient and geometric bridges remain explicit. [pcc:unverified]

Protocol, evidence and limits

Protocol `AUDIT-MUMFORD-CENSUS-20261109-v1` was frozen at 22:33:31Z, hash prefix `a7bc5ba9` (full hash in the protocol and export). It was committed in `996411bd` and published as `d650560d` before the first population call. Arithmetic source is `11100a85`; both binary, compiler, lock and exact upstream MIT-backend pins are retained in `../../native-census-v1/sources.json`. The source and analysis were fixed before data. [pcc:unverified]

Each cell runs Rust census, C++ full recomputation and source-frozen Rust analysis under a 600-second cap. Analysis regenerates the TSV from JSON and requires byte equality. C++ uses separate polynomial arithmetic and direct Fp/Fp^2 point counts as additional genus-two controls. Every output pair must be in the complete enumeration, and the entire finite group table is checked. [pcc:unverified]

Fixed execution order	Metadata-only seed	Diagnostic seconds
F3 r0	202611093001	3.64
F5 r0	202611095001	447.76
F3 r1	202611093002	5.20
F5 r1	202611095002	561.64

Flow: four planned, four attempted, four completed, zero interrupted, zero resumed, zero excluded, zero failed or timed out. No warmups or retries. All cells use one thread on host `762f5fc120c9`, existing local capacity, zero paid compute, isolation L0. All clocks are invalid as scientific timing; no speed claim uses these intervals. The pre-data 660-second total planning estimate was an extrapolation, not an observed population runtime. [pcc:unverified]

The export passes canonical schema/protocol/file verification with zero findings. The archive contains every one of the 40 native/raw/launch files; an unpack-and-hash check matches all 40 to the manifest. Both fields' `raw.jsonl`, `audit.tsv`, C++ result and point-count TSV, `finish.json` and `analysis.json` repeat exactly. Launch/start/seed/time metadata intentionally differs. See `../../evidence/census-v1/` for the archive, manifest, summary files, repeat receipt and unsigned PCC record. [pcc:unverified]

These are exact finite model counts, not a typical-subgroup experiment, a fixed-genus asymptotic limit, a performance measurement, an independent machine check or a formal Picard/Jacobian identification. They do not replace the original all-p-below-500 elliptic census, binary/ternary requests, qualified pairing/solver/walk obligations, or the large anomalous Smart-attack instance. The fourteen-item audit remains partial. [pcc:unverified]

Primary context: Stoll2014 representation/addition sections and Example 4.19, read at the scopes in `../../stoll-sources.json`; Gaudry/MMT source assumptions remain in the main source map. No novelty claim is made. Original Cantor1987 remains unread after publisher403. [pcc:unverified]